Subject: Embedded System Lab

Subject Code: 7IT06

Experiment No. 2

Title: Write and Upload a Simple Program Using C Language on Arduino Uno Board to Display "Hello" over the Serial Monitor

Objectives:

- To introduce the Arduino Uno development board and its programming workflow.
- To learn how to initialize and use serial communication between the Arduino and computer.
- To write a simple program in C to display a message ("Hello") on the Serial Monitor.
- To compile, upload, and execute a program on Arduino Uno using Arduino IDE.
- To observe and verify the output through the Serial Monitor interface.

Required Components/Kit:

- Arduino Uno Board
- USB Cable (Type A-B)
- Computer with Arduino IDE

Requirement Software:

- Arduino IDE (Install from: https://www.arduino.cc/en/software)

Steps to Install Arduino IDE:

1. Download the installer for your OS from the official Arduino website.
2. Run the setup and install necessary drivers.
3. Open Arduino IDE.
4. Connect Arduino Uno using USB cable.
5. Go to:
    - Tools > Board > Arduino Uno
    - Tools > Port > COMx (where x is the port number of your Arduino)

Concept Explanation:

- Arduino Uno is programmed using a simplified version of C/C++ language.
- Serial communication allows data to be printed and viewed on the Serial Monitor, useful for debugging or interaction.
- The command Serial.begin() initializes communication, and Serial.println() prints messages.

Code:

```
void setup()
{
  Serial.begin(9600);      // Initialize serial communication at 9600 baud rate
}

void loop()
{
  Serial.println("Hello"); // Print "Hello" on Serial Monitor
  delay(1000);             // Wait for 1 second
}
```

Steps to Upload the Program:

1. Open Arduino IDE.
2. Copy and paste the above code into the editor.
3. Click on Verify (tick icon) to compile the code.
4. Click Upload (right arrow icon) to upload the code to Arduino Uno.
5. Go to Tools > Serial Monitor or press Ctrl + Shift + M.
6. Ensure baud rate is set to 9600 at the bottom-right of the Serial Monitor.
7. Observe the repeated display of the word "Hello" every second.

Observation Table:

Serial Output	Time Interval	Baud Rate
Hello	1 second	9600

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Hello	1 second	9600
...	...	...

Conclusion:

The program successfully displayed the message "Hello" on the Serial Monitor, demonstrating basic C programming, serial communication, and Arduino Uno interfacing.